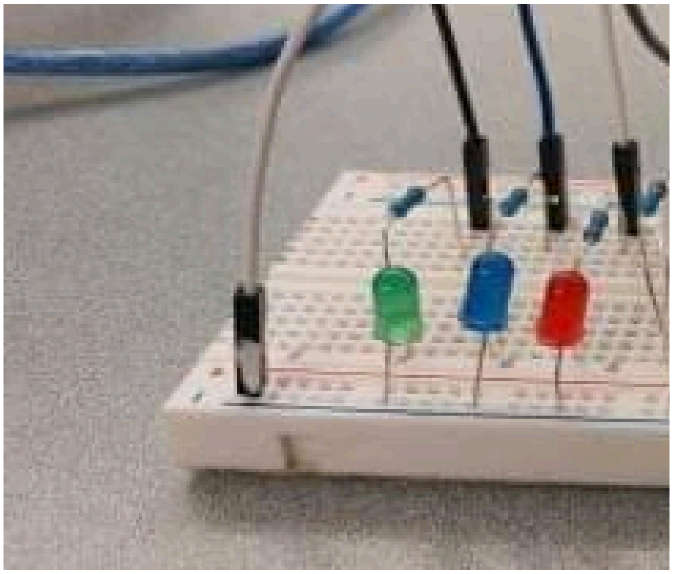
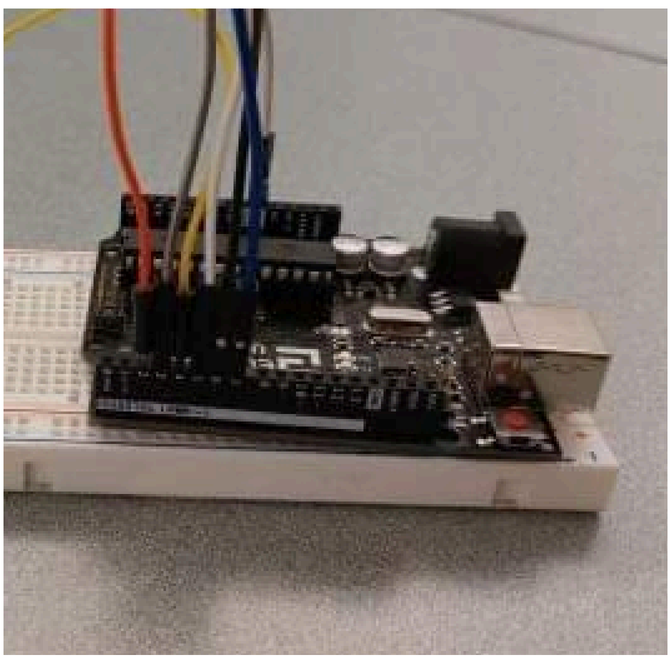
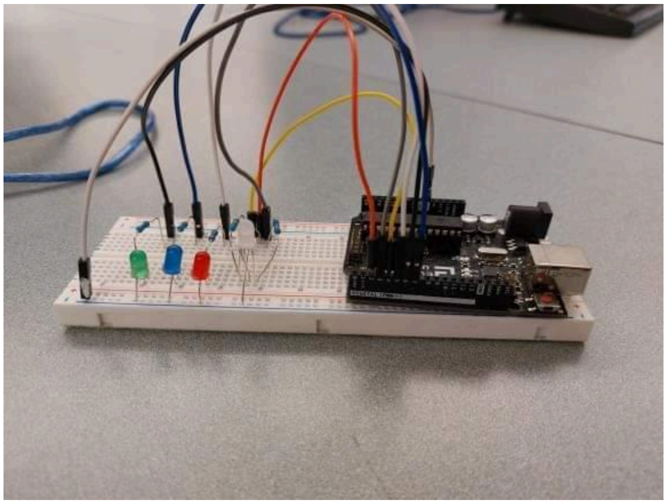
Arduino Engineering Notes

Date	Objectives	What we managed to do	Questions to address
Sept 26	Lessons 1-3 - Learning how LEDs work  	void setup() { // initialize digital pin LED_BUILTIN as an output. for (int i = 4; i < 7; i++) pinMode(i, OUTPUT); } // the loop function runs over and over again forever void loop() { for (int i = 4; i < 7; i++) { digitalWrite(i, HIGH); // turn the LED on (HIGH is the voltage level) delay(100); // wait for a second digitalWrite(i, LOW); // turn the LED off by making the voltage LOW delay(100); } // wait for a second }	Why are there so many grounds, and what does that even mean? - - - During this class, we learned how the breadboard and Arduino board work. As well as that, we learned how the wires operate and how to implement lights so that they turn on. We used resistors to manipulate the voltage going into the lights, so they wouldn't blow.

Sept 27

RGB LEDs



```
int redPin = 4;
int greenPin = 2;
int bluePin = 3;
int number = 0;

//#define COMMON_ANODE

void setup() {
  pinMode(redPin, OUTPUT);
  pinMode(greenPin, OUTPUT);
  pinMode(bluePin, OUTPUT);
}

void loop() {
  number = (int)random(1, 7);
  if (number == 1) {
    setColor(255, 0, 0);  // blue
    delay(1000);
  }
  if (number == 2) {
    setColor(0, 255, 0);  // green
    delay(1000);
  }
  if (number == 3) {
    setColor(0, 0, 255);  // red
    delay(1000);
  }
  if (number == 4) {
    setColor(255, 255, 0);  // aqua
    delay(1000);
  }
  if (number == 5) {
    setColor(150, 0, 80);  // purple
    delay(1000);
  }
  if (number == 6) {
    setColor(0, 255, 255);  // yellow
    delay(1000);
  }
}

void setColor(int red, int green, int blue) {
#ifdef COMMON_ANODE
  red = 255 - red;
  green = 255 - green;
  blue = 255 - blue;
#endif
  analogWrite(redPin, red);
  analogWrite(greenPin, green);
  analogWrite(bluePin, blue);
}
```

What is Common Anode, and why do we need it?

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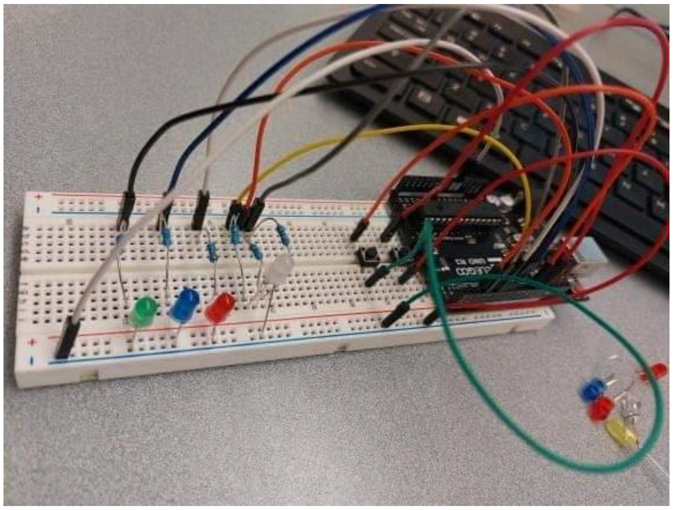
In this class, we used a LED colour changing light where we experimented using different RGB values to produce colours. We discovered that it does not go in RGB order, but instead in BGR which had us confused initially. We also implemented a random number action so that the lights would alternate their colours in a random order, as opposed to the original order they were set up in.

<table>
<tr>
<td>Sept 28</td>
<td>Putting in inputs(buttons)


</td>
<td>

```
int buttonApin = 10;
int buttonBpin = 11;
int redPin = 4;
int greenPin = 2;
int bluePin = 3;
int number = 0;

void setup() {
  // initialize digital pin LED_BUILTIN as an output.
  for (int i = 2; i < 8; i++)
    pinMode(i, OUTPUT);
  pinMode(redPin, OUTPUT);
  pinMode(greenPin, OUTPUT);
  pinMode(bluePin, OUTPUT);
  Serial.begin(9600);
  while (!Serial)
    ;
  Serial.println("started");
  pinMode(buttonApin, INPUT_PULLUP);
  pinMode(buttonBpin, INPUT_PULLUP);
}

void loop() {

  for (int i = 2; i < 8; i++) {
    {
      if (digitalRead(buttonApin) == LOW) {
        digitalWrite(i, HIGH);
      }
    }
    for (int i = 2; i < 12; i++) {
      if (digitalRead(buttonBpin) == LOW) {
        digitalWrite(i, LOW);
      }
    }
    // digitalWrite(i, HIGH);
    //delay(400);
    //digitalWrite(i, LOW);
    // delay(400);
    //Serial.print("LED ");
    //Serial.println(i);
  }
  number = (int)random(1, 7);
  if (number == 1) {
    setColor(255, 0, 0);  // blue
    delay(1000);
  }
  if (number == 2) {
    setColor(0, 255, 0);  // green
    delay(1000);
  }
  if (number == 3) {
    setColor(0, 0, 255);  // red
    delay(1000);
  }
  if (number == 4) {
    setColor(255, 255, 0);  // aqua
    delay(1000);
  }
  if (number == 5) {
    setColor(150, 0, 80);  // purple
    delay(1000);
  }
  if (number == 6) {
    setColor(0, 255, 255);  // yellow
    delay(1000);
  }
```

</td>
<td>Why are the buttons not working? Do the wires need to be specific?
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In this class we had tried to get the buttons to turn the lights on and off, but we were unsuccessful. We believe it may have had something to do with the LED light, but we never tested that theory.</td>
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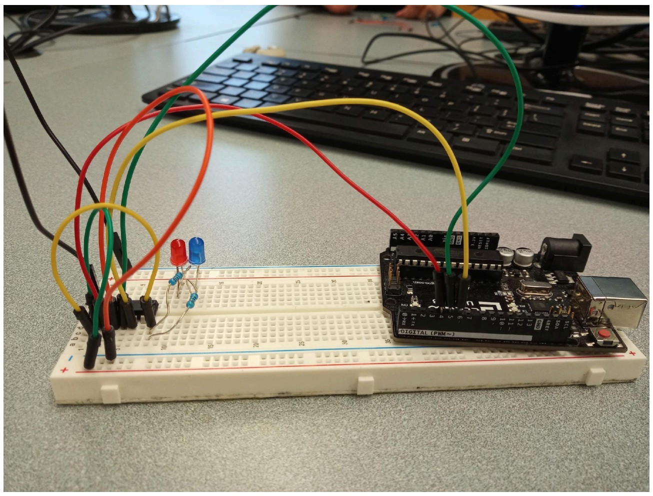
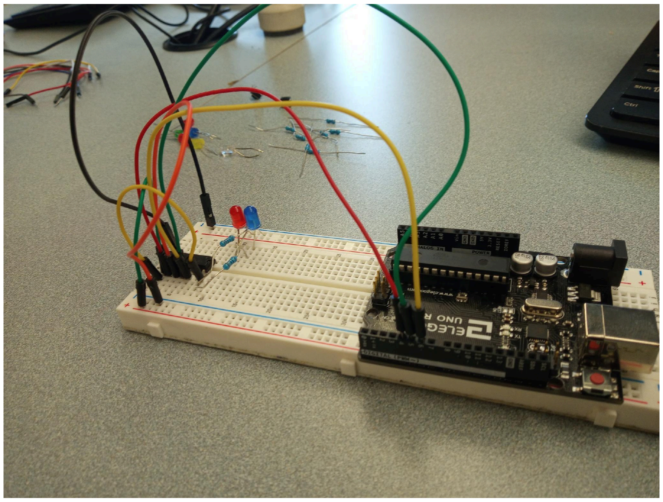
		} void setColor(int red, int green, int blue) { #ifdef COMMON_ANODE red = 255 - red; green = 255 - green; blue = 255 - blue; #endif analogWrite(redPin, red); analogWrite(greenPin, green); analogWrite(bluePin, blue); }	
Sept 29	Eight LEDs and Shift register 	int latchPin = 5; int clockPin = 6; int dataPin = 4; byte leds = 0; void setup() { pinMode(latchPin, OUTPUT); pinMode(dataPin, OUTPUT); pinMode(clockPin, OUTPUT); } void loop() { leds = 0; updateShiftRegister(); delay(500); for (int i = 0; i < 8; i++) { bitSet(leds, i); updateShiftRegister(); delay(500); } } void updateShiftRegister() { digitalWrite(latchPin, LOW); shiftOut(dataPin, clockPin, LSBFIRST, leds); digitalWrite(latchPin, HIGH); }	What is a shift register? Where do the wires go and connect? How do resistors work with it? - - - During this class, we learned about the shift register and how it can be used. Unfortunately, following along was difficult and we didn't get much done. We tried to follow some instructions online as well but it did not work out. We tried to follow some instructions online as well but it did not work out. We used the whole class to try and figure out how to turn the two lights on.

<table>
<tr>
<td>Oct 2</td>
<td>Figured out how the button works but it is currently not working and trying to figure out shift register


</td>
<td>

```
int latchPin = 12;
int clockPin = 13;
int dataPin = 11;
int ledPin = 4;
int buttonApin = 9;
int buttonBpin = 8;
byte leds = 0;

void setup() {
  pinMode(latchPin, OUTPUT);
  pinMode(dataPin, OUTPUT);
  pinMode(clockPin, OUTPUT);
  Serial.begin(9600);
}

void loop() {
 // if (digitalRead(buttonApin) == LOW) {
    //digitalWrite(ledPin, HIGH);
    //Serial.println("Adown");
  //}
  //if (digitalRead(buttonBpin) == LOW) {
 //   digitalWrite(ledPin, LOW);
 //   Serial.println("Bdown");
 // }
  leds = 0;
  updateShiftRegister();
  for (int i = 0; i < 8; i++) {
    bitSet(leds, i);
    updateShiftRegister();
  }
}

void updateShiftRegister() {
  digitalWrite(latchPin, LOW);
  shiftOut(dataPin, clockPin, LSBFIRST,
leds);
  digitalWrite(latchPin, HIGH);
}
```

</td>
<td>The buttons used to work, but not with the shift register?
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Finally we managed to understand how the buttons worked, where the wires had to specifically follow a diagonal pattern. Unfortunately, once we introduced the shift register, it had stopped working. Instead of trying to get it to work, we focused on getting the lights to work with the shift register first, and then we focused a bit on the buttons.</td>
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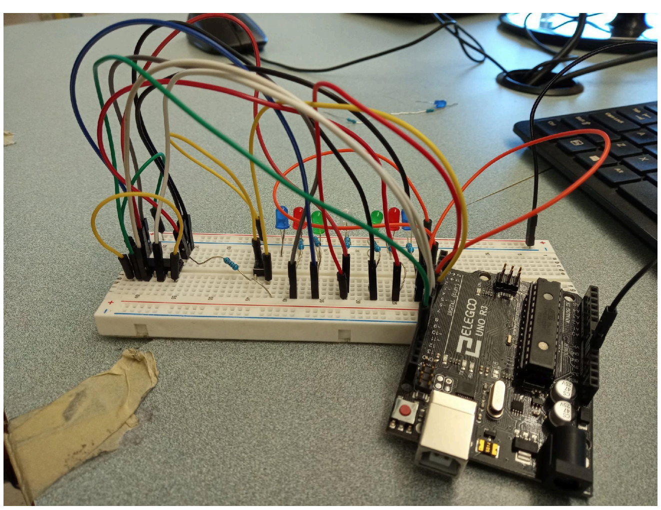
<td>Oct 3</td>
<td>tug-of-war day 1


</td>
<td>

```
int latchPin = 12;
int clockPin = 13;
int dataPin = 11;
int ledPin = 4;
int buttonApin = 2;
int buttonBpin = 3;
int button = 1;
byte leds = 0;

void setup() {
  pinMode(latchPin, OUTPUT);
  pinMode(dataPin, OUTPUT);
  pinMode(clockPin, OUTPUT);
  pinMode(buttonApin, INPUT_PULLUP);
  pinMode(buttonBpin, INPUT_PULLUP);
  Serial.begin(9600);
  leds = 0;
  updateShiftRegister();
}

void loop() {
  bitSet(leds, button);
  updateShiftRegister();
  if (digitalRead(buttonApin) == LOW) {
```

</td>
<td>Where do we start? How do we only get one light on? How do we do the rest of the game?
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During this class, we set the whole layout of the game so we knew what needed to be done. We managed to get the shift</td>
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<td></td>
<td></td>
<td>button += 1;
Serial.println("A");
checkEnd();
}
if (digitalRead(buttonBpin) == LOW) {
button -= 1;
Serial.println("B");
checkEnd();
}
}
void updateShiftRegister() {
digitalWrite(latchPin, LOW);
shiftOut(dataPin, clockPin, LSBFIRST, leds);
digitalWrite(latchPin, HIGH);
}</td>
<td>register to work with the whole set of 7 lights, so it was time to introduce the button component to get the lights on and off. Eventually we realized it would be better to instead focus on getting one light to move, so we attempted to solve that issue. Although the light that lit up was not the one we wanted, we figured out a way for it to work. Then we tried to implement the button code, and it registered but did not do what we wanted it to do.</td>
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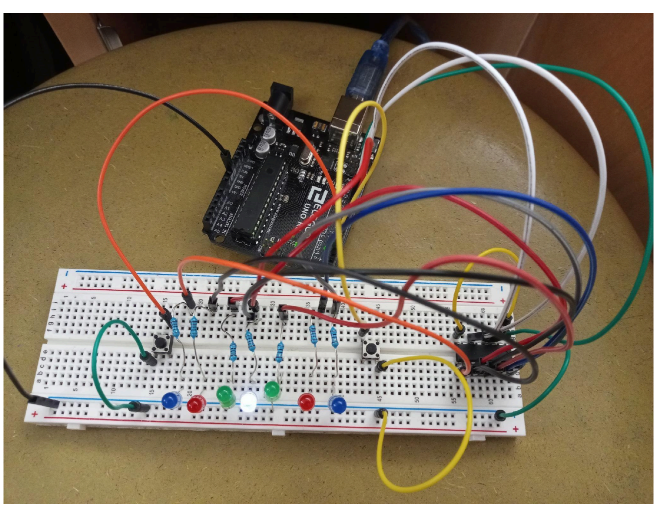
<td>Oct 4</td>
<td>tug-of-war day 2
</td>
<td>int latchPin = 12;
int clockPin = 13;
int dataPin = 11;
int ledPin = 4;
int buttonApin = 2;
int buttonBpin = 3;
int button = 3;
int end = 0;
byte leds = 0;

void setup() {
pinMode(latchPin, OUTPUT);
pinMode(dataPin, OUTPUT);
pinMode(clockPin, OUTPUT);
pinMode(buttonApin, INPUT_PULLUP);
pinMode(buttonBpin, INPUT_PULLUP);
Serial.begin(9600);
leds = 0;
}

void loop() {
bitSet(leds, button);
updateShiftRegister();
if (digitalRead(buttonApin) == LOW ||</td>
<td>Celebration time for the game, should we differentiate it?
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We had finally let the light in the middle be the real starting point. We noticed the buttons were always being read as pushed, even if it was one click, so we</td>
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</table>

```
digitalRead(buttonBpin) == LOW) {
    if (digitalRead(buttonApin) == LOW) {
      button += 1;
      leds = 1;
      bitSet(leds, button);
      updateShiftRegister();
      Serial.println("A");
      Serial.println(button);
      leds = 0;
      bitSet(leds, button);
      updateShiftRegister();
      while (digitalRead(buttonApin) ==
LOW || digitalRead(buttonBpin) == LOW)
        ;
      checkEnd();
    }
    if (digitalRead(buttonBpin) == LOW) {
      button -= 1;
      leds = 1;
      bitSet(leds, button);
      updateShiftRegister();
      Serial.println("B");
      Serial.println(button);
      leds = 0;
      bitSet(leds, button);
      updateShiftRegister();
      while (digitalRead(buttonApin) ==
LOW || digitalRead(buttonBpin) == LOW)
        ;
      checkEnd();
    }
  }
}

void updateShiftRegister() {
  digitalWrite(latchPin, LOW);
  shiftOut(dataPin, clockPin, LSBFIRST,
leds);
  digitalWrite(latchPin, HIGH);
}

void checkEnd() {
  if (button == 0 || button == 6) {
    if (button == 0) {
      for (int i = 0; i < 8; i++) {
        leds = 1;
        bitSet(leds, i);
        updateShiftRegister();
        delay(800);
        leds = 0;
        bitSet(leds, i);
        updateShiftRegister();
        end = 1;
      }
    }
    if (button == 6) {
      for (int i = 7; i > 0; i--) {
        leds = 1;
        bitSet(leds, i);
        updateShiftRegister();
        delay(800);
        leds = 0;
        bitSet(leds, i);
        updateShiftRegister();
        end = 1;
      }
    }
  }
```

used delay to stop this. We realized this hindered the game play by interrupting a push. We were then able to introduce a while command to ensure the button values were being read accurately, and the game was still playable. All that was left to do was to create a celebration event using a checkEnd function. This task was quick, however we realized it was difficult to tell who had won if the animation was the same. Eventually, we had the lights following a pattern starting from the side that won, and then restarting the game.

		`` if (end == 1) restart(); else loop(); } void restart() { button = 3; end = 0; loop(); } ```	